

ONLINE BANKING SYSTEM

PROJECT REPORT

Technologies Used

Java Swing, JDBC and MySQL

Submitted By

Simran Joshi

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1. Introduction

The Online Banking System is a desktop-based banking application developed using Java Swing, JDBC, and MySQL. The application provides users with essential banking functionalities such as registration, login, account creation, deposits, withdrawals, fund transfers, and transaction history management through an interactive graphical user interface.

The project aims to demonstrate the implementation of database connectivity, transaction processing, and user-friendly software design using Java technologies. The system securely stores and manages banking information using a MySQL database and enables users to perform banking operations efficiently.

The application follows a modular structure consisting of graphical user interface components, data access objects (DAO), database connectivity modules, and transaction management functionalities. This architecture ensures maintainability, scalability, and efficient interaction between the user interface and database.

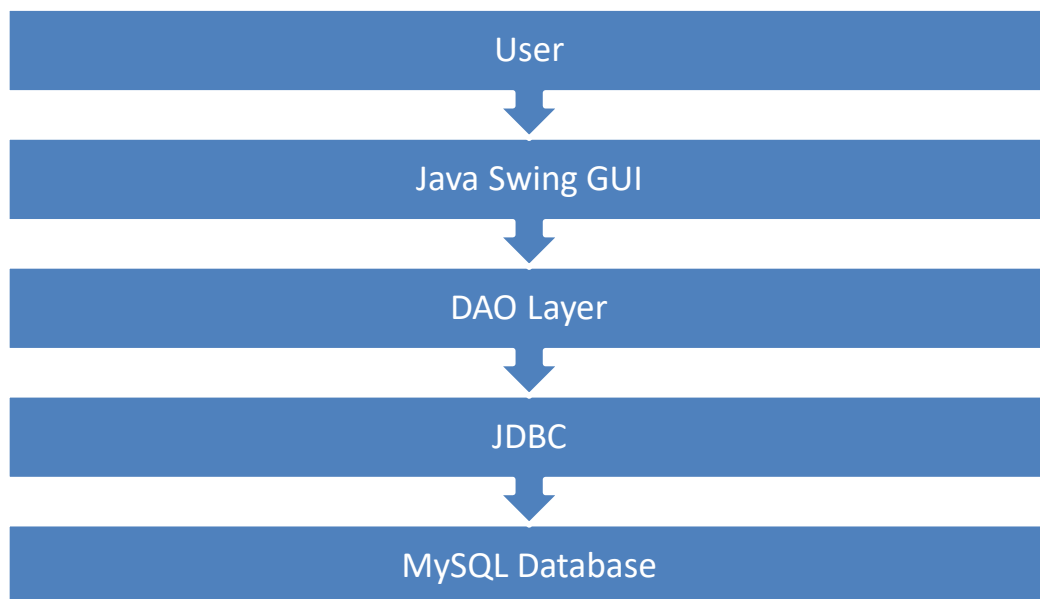
2. Objectives

- To develop a banking application using Java.
- To implement a graphical user interface using Java Swing.
- To establish connectivity between Java and MySQL using JDBC.
- To provide secure user registration and authentication.
- To manage customer accounts efficiently.
- To perform deposit, withdrawal, and fund transfer operations.
- To maintain transaction history records.
- To provide a user-friendly banking experience.

3. Technologies Used

Technology	Purpose
Java	Core Application Development
Swing	Graphical User Interface Development
JDBC	Database Connectivity
MySQL	Database Management
VS Code	Code Development and Testing

4. System Architecture



The system follows a layered architecture where users interact with the Java Swing graphical interface. The GUI communicates with the DAO layer, which contains the business logic and database operations. JDBC is used to establish communication with the MySQL database for storing and retrieving banking information.

5. Database Design

The Online Banking System uses MySQL as its backend database. The database is designed to store user information, account details, and transaction records. Three main tables were created to manage banking operations efficiently.

Users Table

Field Name	Data Type	Description
user_id	INT	Unique User ID
username	VARCHAR	Username
password	VARCHAR	User Password
full_name	VARCHAR	Full Name

Accounts Table

Field Name	Data Type	Description
account_id	INT	Unique Account ID
user_id	INT	Linked User ID
account_number	VARCHAR	Account Number
balance	DOUBLE	Account Balance

Transactions Table

Field Name	Data Type	Description
transaction_id	INT	Unique Transaction ID
account_id	INT	Linked Account ID
transaction_type	VARCHAR	Deposit/Withdraw
amount	DOUBLE	Transaction Amount
transaction_date	TIMESTAMP	Transaction Date

6. Modules Implemented

1. User Registration Module

- This module allows new users to create an account by providing their username, password, and full name. The information is stored securely in the MySQL database.

2. User Login Module

- This module authenticates users by verifying their username and password against the records stored in the database.

3. Account Management Module

- This module enables users to create bank accounts and maintain account information.

4. Deposit Module

- This module allows users to deposit money into their accounts and automatically records the transaction.

5. Withdrawal Module

- This module allows users to withdraw funds from their accounts while validating available balance.

6. Fund Transfer Module

- This module enables users to transfer money from one account to another.

7. Transaction History Module

- This module displays transaction records stored in the database.

7. GUI Screens

Figure 1: Login Screen

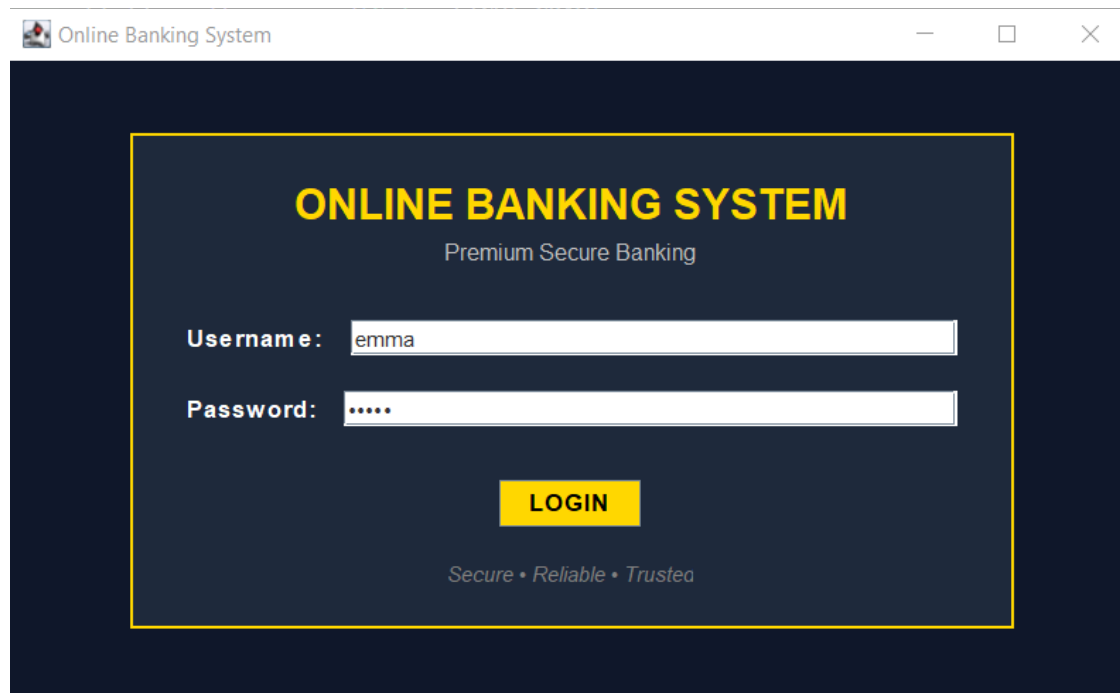


Figure 1 shows the login screen of the Online Banking System.

Figure 2: Dashboard Screen

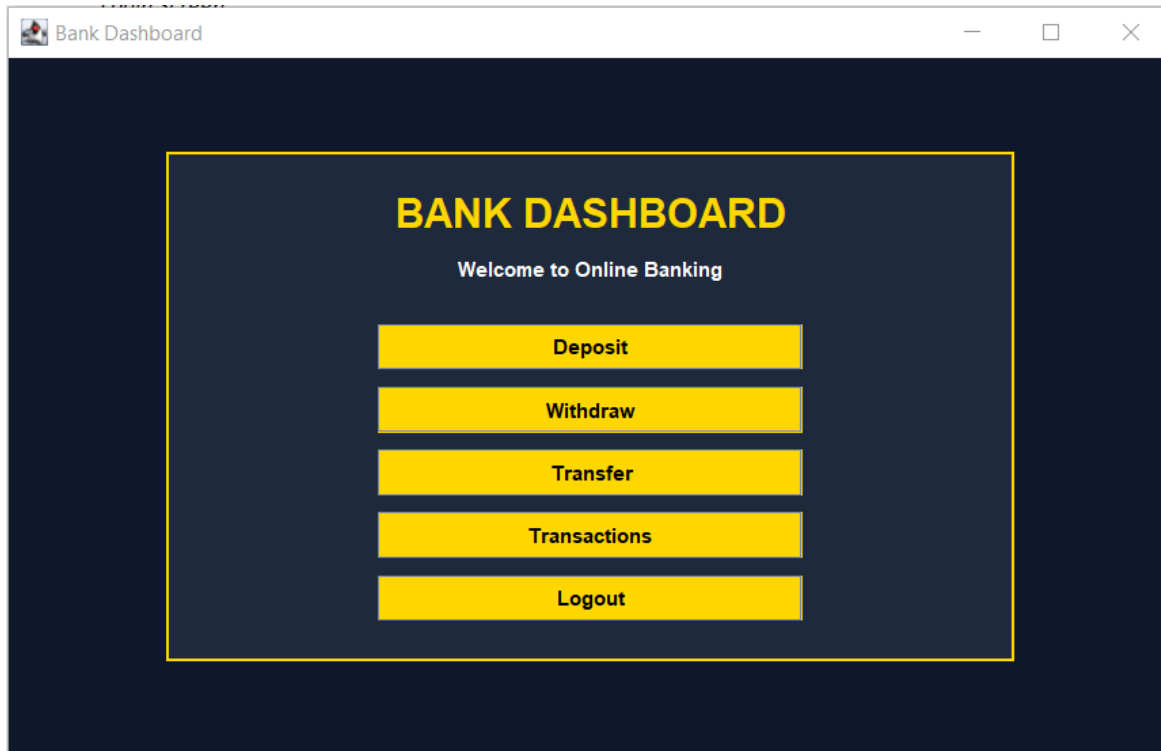


Figure 2 shows the dashboard screen of the Online Banking System.

Figure 3: Deposit Window

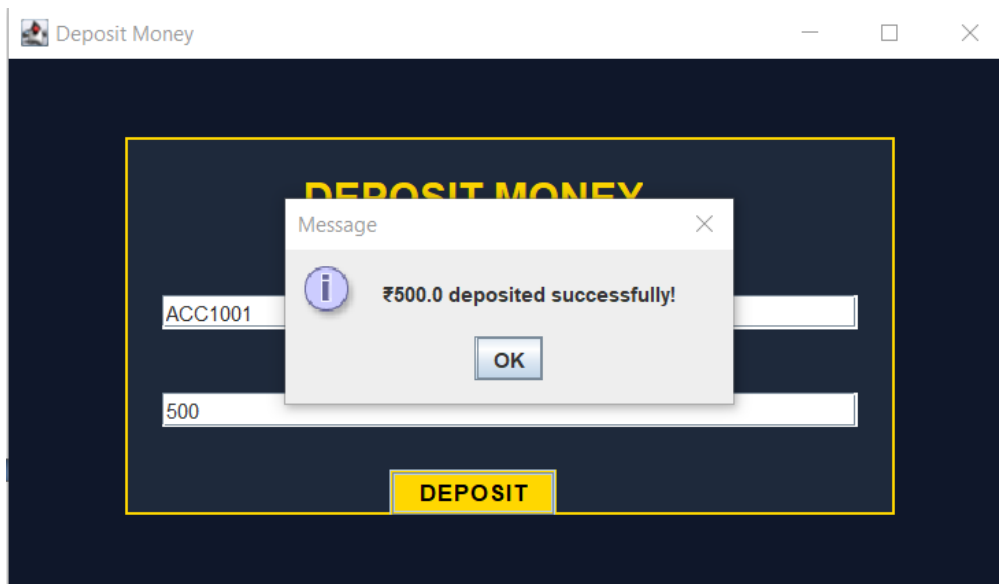
Before Depositing Money

```
35 • SELECT * FROM accounts;
36 • SELECT * FROM transactions;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: [TA](#)

	account_id	user_id	account_number	balance
▶	1	1	ACC1001	400.00
	2	1	ACC1002	6200.00
	3	2	ACC2001	12400.00
*	NULL	NULL	NULL	NULL

Action of Depositing Money



After Depositing Money

```
35 • SELECT * FROM accounts;
36 • SELECT * FROM transactions;
```

	account_id	user_id	account_number	balance
▶	1	1	ACC1001	900.00
	2	1	ACC1002	6200.00
	3	2	ACC2001	12400.00
*	NULL	NULL	NULL	NULL

Figure 4: Withdraw Window

Before Withdrawing Money

```
35 • SELECT * FROM accounts;
36 • SELECT * FROM transactions;
```

Result Grid | Filter Rows: | Edit:

account_id	user_id	account_number	balance
1	1	ACC1001	900.00
2	1	ACC1002	6200.00
3	2	ACC2001	12400.00
NULL	NULL	NULL	NULL

Action of Withdrawing Money

Withdraw Money

WITHDRAW MONEY

ACC1001

200

WITHDRAW

Message

₹200.0 withdrawn successfully!

OK

After Withdrawing Money

```
35 • SELECT * FROM accounts;
36 • SELECT * FROM transactions;
```

< | Result Grid | Filter Rows: | Edit:

	account_id	user_id	account_number	balance
▶	1	1	ACC1001	700.00
	2	1	ACC1002	6200.00
	3	2	ACC2001	12400.00
*	NULL	NULL	NULL	NULL

Figure 5: Transfer Window

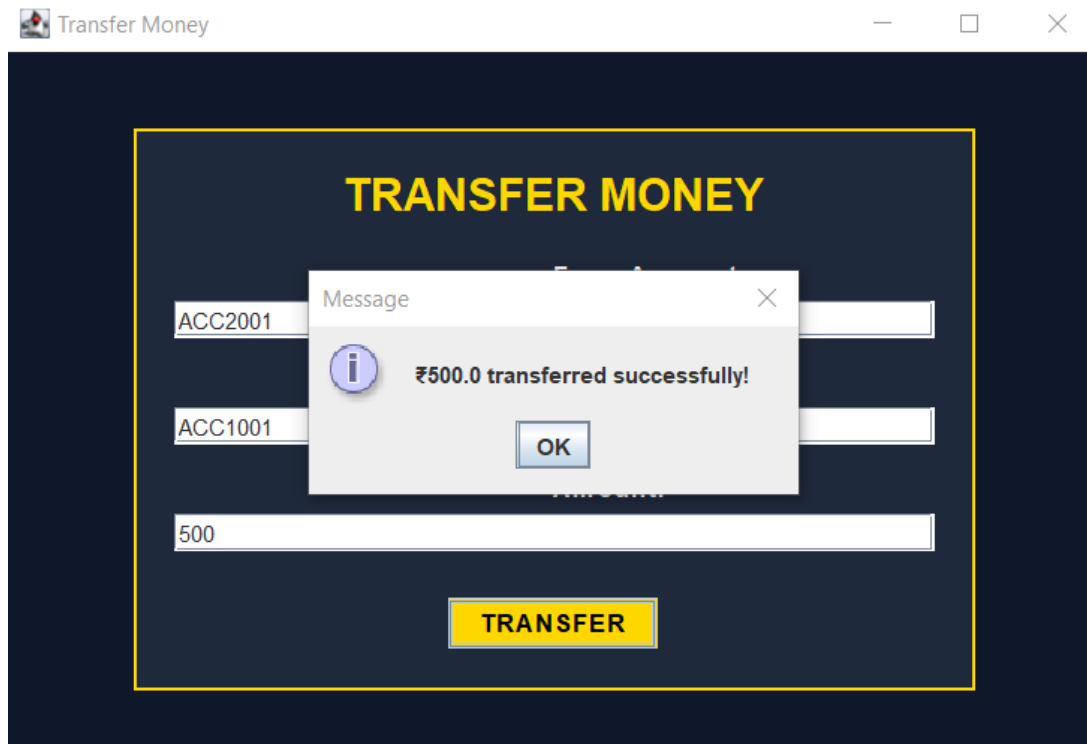
Before Transferring Money

```
35 • SELECT * FROM accounts;
36 • SELECT * FROM transactions;
```

< | Result Grid | Filter Rows: | Edit:

	account_id	user_id	account_number	balance
▶	1	1	ACC1001	700.00
	2	1	ACC1002	6200.00
	3	2	ACC2001	12400.00
*	NULL	NULL	NULL	NULL

Action of Transferring Money



After Transferring Money

```
35 • SELECT * FROM accounts;
36 • SELECT * FROM transactions;
```

Result Grid | Filter Rows: | Edit: |

	account_id	user_id	account_number	balance
▶	1	1	ACC1001	1200.00
	2	1	ACC1002	6200.00
	3	2	ACC2001	11900.00
*	NULL	NULL	NULL	NULL

Figure 6: Transaction History Window

TRANSACTION HISTORY				
ID	Account ID	Type	Amount	Date
1	1	DEPOSIT	5000.0	2026-06-05 23:39:46.0
2	3	DEPOSIT	500.0	2026-06-06 20:58:54.0
3	3	DEPOSIT	1500.0	2026-06-06 20:59:58.0
4	3	WITHDRAW	100.0	2026-06-06 21:02:59.0
6	3	TRANSFER_IN	500.0	2026-06-06 21:07:43.0
9	1	DEPOSIT	100.0	2026-06-06 21:37:08.0
10	1	DEPOSIT	100.0	2026-06-06 21:41:10.0
11	3	WITHDRAW	100.0	2026-06-06 23:38:57.0
12	2	DEPOSIT	5000.0	2026-06-06 23:46:24.0
13	2	WITHDRAW	1000.0	2026-06-06 23:46:45.0
14	1	DEPOSIT	500.0	2026-06-07 00:28:49.0
15	1	WITHDRAW	200.0	2026-06-07 00:35:05.0

9. Results

The Online Banking System was successfully developed and tested using Java Swing, JDBC, and MySQL. The application successfully performed all intended banking operations through a graphical user interface.

The following functionalities were implemented and verified:

- **User Registration** – New users were successfully registered and stored in the database.
- **User Login** – Users were authenticated using their username and password.
- **Account Creation** – Bank accounts were created and linked to registered users.
- **Deposit Operation** – Users were able to deposit funds into their accounts, and balances were updated correctly.
- **Withdrawal Operation** – Users were able to withdraw funds from their accounts while maintaining balance validation.
- **Fund Transfer** – Money was successfully transferred between accounts.
- **Transaction History** – Transaction records were stored and displayed successfully.

- **Database Integration** – All data was stored and retrieved from the MySQL database using JDBC connectivity.

The test results confirmed that the application performed all banking operations accurately and efficiently.

10. Future Enhancements

Although the current version of the Online Banking System successfully performs essential banking operations, several enhancements can be implemented in future versions to improve functionality and security.

Future improvements may include:

- Password encryption for enhanced security.
- User profile management and account settings.
- Account balance dashboard with graphical reports.
- Email and SMS notifications for transactions.
- Admin panel for managing users and accounts.
- Online bill payment functionality.
- Loan and fixed deposit management modules.
- Mobile application integration.
- Multi-user role management system.

These enhancements would make the application more secure, scalable, and suitable for real-world banking environments.

11. CONCLUSION

The Online Banking System was successfully designed and implemented using Java Swing, JDBC, and MySQL. The project provided practical experience in database management, graphical user interface development, and transaction processing.

The application enables users to perform essential banking operations such as registration, login, account creation, deposits, withdrawals, fund transfers, and transaction history management through a user-friendly interface.

The project demonstrates the effective integration of Java technologies with a relational database system and provides a strong foundation for understanding real-world banking application development. Overall, the project successfully achieved its objectives and delivered a functional and reliable banking management system.